

İREM KARACA

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EDUCATION

- M.Sc. Machine Learning** Oct 2024 - Present
Eberhard Karls University of Tübingen, *Tübingen, Germany*
 - ELIZA research-oriented master's scholar
- B.Sc. Computer Engineering** Sep 2019 - Jun 2024
Koç University, College of Engineering, *Istanbul, Türkiye*
 - GPA: 3.75/4, Summa Cum Laude

WORK EXPERIENCE

- Research Assistant** Feb 2025 - Present
ELLIS Institute Tübingen, *Tübingen, Germany*
 - Developing student simulation models for the AI Tutor project supervised by Dr. Wieland Brendel.
- Research Intern** Jul - Sep 2023
Brussels Human Robotics Research Center - Vrije Universiteit Brussel, *Brussels, Belgium*
 - Developed machine learning models to estimate muscle force for upper-limb exoskeletons using ultrasound measurements, contributing to advancements in assistive robotics.
- Teaching Assistant** Feb 2022 - Feb 2023
Koç University, *Istanbul, Türkiye*
 - Conducted office hours and lab sessions for the "Computer Systems & Programming" course, providing academic support to students and assisting with programming exercises.
- Computer Vision Engineer Intern** Jun - Sep 2022
Delivers.AI, *Istanbul, Türkiye*
 - Worked on object detection for an autonomous delivery robot, improving the performance of a machine learning model using YOLOv4, achieving a higher mAP compared to the previous model.
- Software Development Intern** Jul - Aug 2021
OBSS, *Istanbul, Türkiye*
 - Developed a full-stack online marketplace web application using Java, Spring Boot, React, and MySQL, enhancing user experience and functionality.
- Research Assistant** Nov 2019 - Nov 2020
Intelligent User Interfaces Laboratory - Koç University, *Istanbul, Türkiye*
 - Assisted graduate students in research on sketch-based interfaces and their applications in creative processes.

FELLOWSHIP/MENTEE EXPERIENCE

- Master's Student Fellow** Oct 2024 - Present
Konrad Zuse School of Excellence in Learning and Intelligent Systems (ELIZA), *Tübingen, Germany*
 - Awarded a scholarship and supervision through ELIZA, in collaboration with the European Laboratory for Learning and Intelligent Systems (ELLIS), as part of my master's program in machine learning.
- Machine Learning Bootcamp Participant** May - Oct 2023
Google Developers Group & inzva, *Remote*
 - Participated in a 5-month bootcamp that included Coursera courses and weekly classes, focusing on core machine learning concepts and practical applications.
- Mentee** Oct 2022 - Mar 2023
Transcend - Bending Spoons, *Remote*
 - Participated in a six-month one-on-one mentorship with a senior engineer at Bending Spoons, focusing on professional growth and technical skill development.
- Mentee** Sep 2021 - Apr 2022
inzva, *Remote*
 - Received mentorship from a senior engineer and inzva alumna for eight months, gaining valuable insights into the tech industry and professional development.
- Young Women Leaders Fellowship** Jun 2021 - Apr 2022
Yenibirlider (A-new-leader) Association & Turkish Confederation of Employer Associations, *Istanbul, Türkiye*
 - Completed training and participated in seminars on leadership, career management, project management, social skills, case studies, and future job trends, focusing on personal and professional growth.

VOLUNTEER EXPERIENCE

Mentorship Program Coordinator

Aug 2022 - Aug 2024

inzva, *Istanbul, Türkiye*

- Coordinated a mentorship program for undergraduate computer science students, connecting them with young professionals from the Turkish tech community to support their academic and career growth.

Volunteer

Jul 2024

Nesin Mathematics Village, *Izmir, Türkiye*

- Assisted in kitchen and library operations at the Nesin Mathematics Village, an educational institute focused on mathematics, supporting daily activities.

Incoming Exchange Students Mentor

Aug 2021 - Jan 2022

Koç University, *Istanbul, Türkiye*

- Supported 10 exchange students in adapting to life in Türkiye and at Koç University by providing guidance and assisting with cultural and academic integration.

Section Leader

Apr - May 2021

Stanford University - Code in Place, *Remote*

- Conducted problem-solving sessions with 10 students from diverse backgrounds and ages, teaching introductory Python programming skills over a 5-week course.

Tutor

Mar - May 2021

Society of Women Engineers Koç University - Girls Who Code, *Remote*

- Tutored female high school students in the Introduction to Python program, helping them develop foundational programming skills.

Section Leader

Feb - Apr 2021

Koç University - CS 101: Hello, World!, *Remote*

- Delivered 75-minute Python lectures to 12 high school students across Türkiye in an 8-week online course, promoting understanding of core programming concepts.

Corporate Relations Manager

Aug 2020 - Jan 2021

Koç University Association for Computing Machinery Student Chapter, *Istanbul, Türkiye*

- Organized technology and career events in collaboration with industry experts, while managing a 15-person team to ensure successful execution.

CERTIFICATES

- **IELTS Score: 7.5** (February 2024)
- **TensorFlow Developer Certificate** - Google (November 2023)
- **30th IEEE Conference on Signal Processing and Communication Applications (SIU 2022)** - IEEE (May 2022)
- **Algorithm Competition Winter Camp 2022** - inzva (February 2022)

LANGUAGES

- **Turkish** - Mother Tongue
- **English** - Fluent
- **German** - Intermediate